

EC571 Fall 2026
Cadence Virtuoso Tutorial I

Acknowledgement

The following individuals have contributed to developing this tutorial: Zeynep Ece Kizilates, Alperen Yasar, Saraja Kadambari, Laith Shamieh, Timur Zirtiloglu, Akshaya Bali, Ajay Joshi, and Rabia Yazicigil.

Lab Description

In this first lab, you will setup your FreePDK45 technical library in the Cadence Virtuoso environment, learn how to create libraries and schematics, and add/edit instances on the schematic. You will also perform DC simulations to obtain I/V curves of the NMOS and PMOS devices.

Start the Desktop Session using the instructions provided in the “Setting up SCC OnDemand Desktop” tutorial.

Setting up access to FreePDK45 library

1. Launch a terminal window. Create a directory “EC571” in your home directory for the class. In that directory, create another directory called “FreePDK45”. This will be your working directory, where you will save your working libraries and use to start the Cadence Virtuoso tool. The commands for completing the steps above are listed below.

```
mkdir EC571
cd EC571
mkdir FreePDK45
cd FreePDK45
```

2. Copy all files from folder “/ad/eng/courses/ec/ec571/Student/FreePDK45” to your FreePDK45 directory. Make sure all hidden files are also copied to your FreePDK45 directory.

```
cp -R /ad/eng/courses/ec/ec571/Student/FreePDK45/* .
```

3. Check if you have got the following files in your FreePDK45 directory: .cdsenv (hidden file), .cdsinit (hidden file), display.drf, cds.lib, lib.defs, run_centos.sh and run_cadence.sh. These files will be used to initialize your working environment. Also, make sure you have “custom models” folders copied.

Opening the Cadence Tool

1. In the FreePDK45 directory, run the following command to start the Cadence with the FreePDK45 design kit:

```
./run_cadence.sh
```

2. After you type the command, the “Command Interpreter Window” (“CIW”) (see Figure 1) will pop up. This is your main Cadence window.

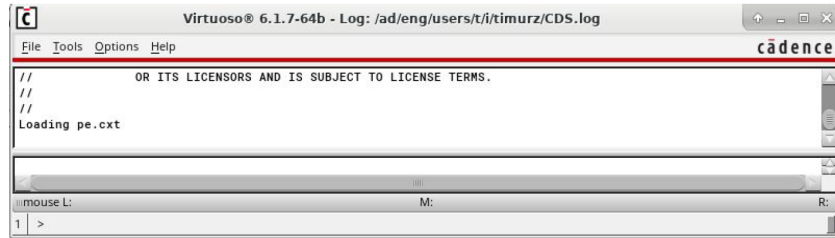


Figure 1: “Command Interpreter Window” (“CIW”)

Create and Access Libraries

1. To create a new library: click File → New → Library (see Figure 2), and name it “EC571_Lab1” for your lab 1. Make sure to select “Attach to an existing technology library” under Technology File, then click “Apply” (see Figure 3).

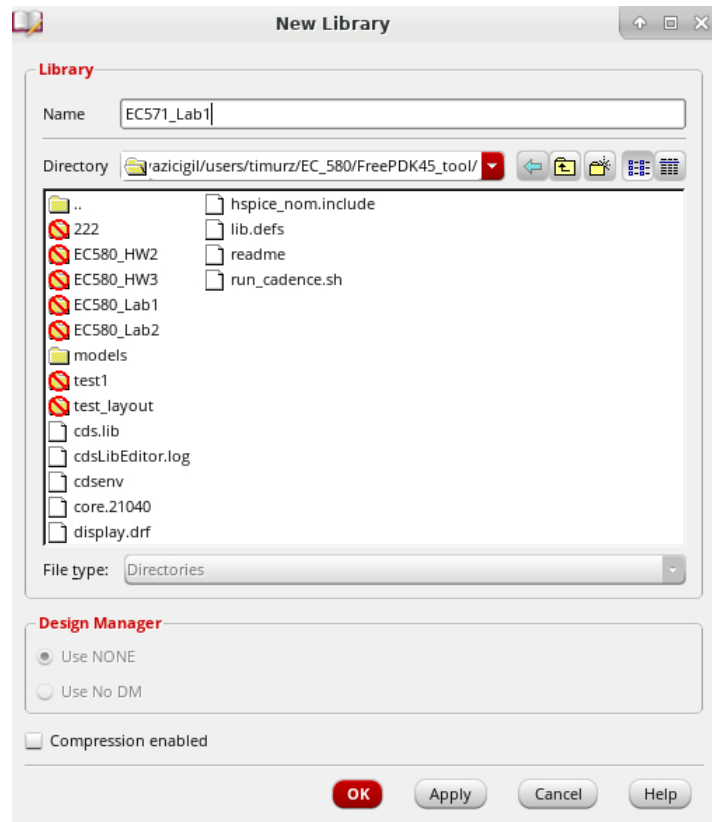


Figure 2: “New library” window

2. In the “Attach Library to Technology Library” window, select “NCSU_techLib_FreePDK45” as your referring library and click “OK” (see Figure 4).
3. To access existing libraries: click Tools → Library Manager (Figure 5). “Library Manager” will open (Figure 6).

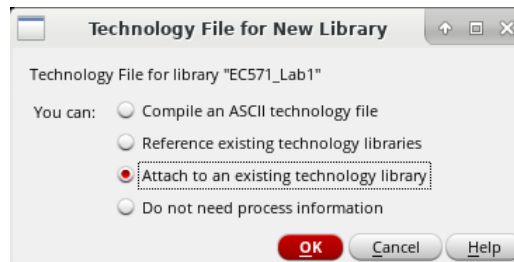


Figure 3: Technology file selection option for new library

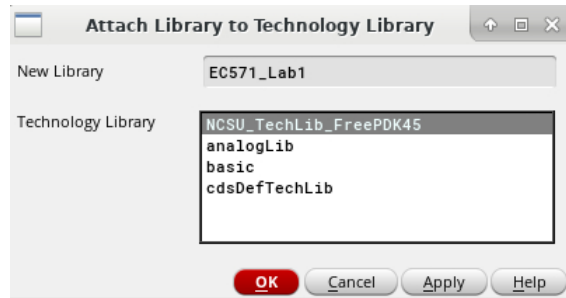


Figure 4: Technology file selection option for new library

Create Schematic

1. In the "Library Manager", select your library, then click File → New → Cell View (Figure 7). Give Cell the name "lab_nmos". Make sure "Type" is schematic, and you are creating this cell view under the right library. Click "OK" (Figure 8).

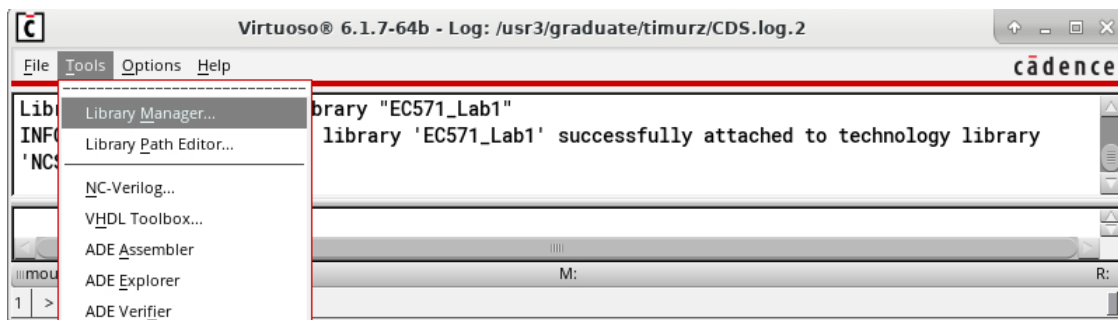


Figure 5: Opening “Library Manager”

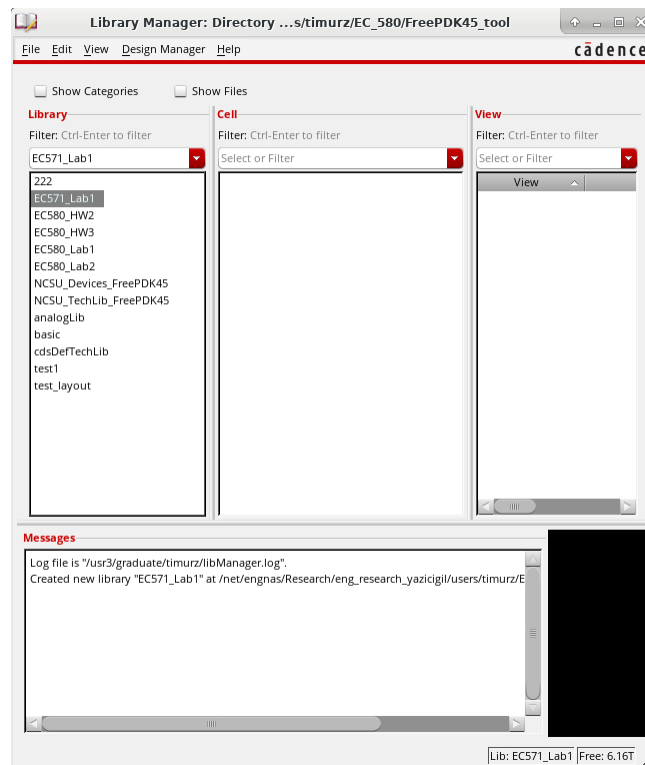


Figure 6: “Library Manager” window

2. A Warning will appear, click on “Yes”, and the “Virtuoso Schematic Editor” window will be opened (Figure 9). This is the the window in which you will be creating and modifying the schematics of the circuits. Don’t worry about the Warning at this point.

Create Schematic View

In this section, we will create schematic view for NMOS transistor simulation as shown on (Figure 10) and simulate it to generate its I/V curve.

1. To add instances for any component, click Create → Instance or just press “i” from the keyboard.
2. Click “Browse” next to Library to open the Library Browser. In the Library Browser Window,

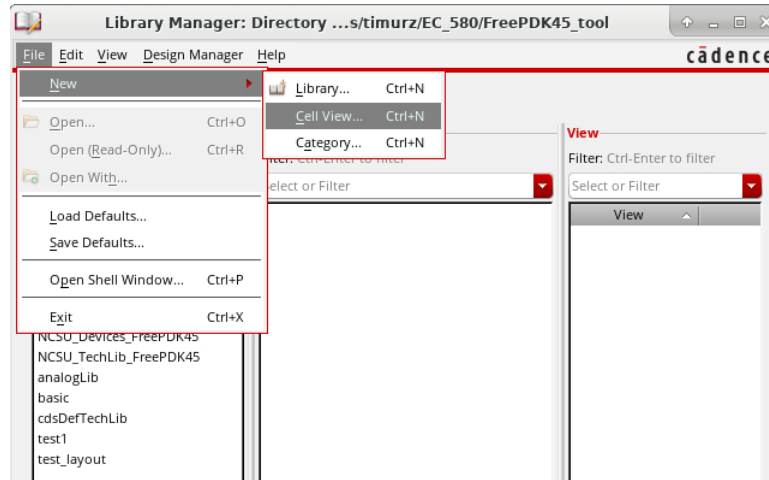


Figure 7: Creating new cell form the “Library manager”

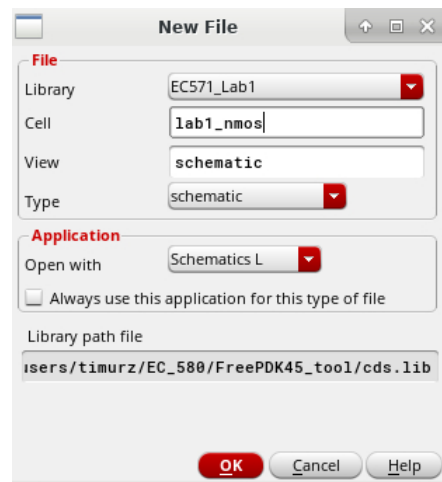


Figure 8: New cell view creation window

choose “NCSU\Devices\FreePDK45” for the library, “NMOS_VTG” for the cell, and “symbol” for the view (Figure 11).

3. For this lab, in the “Add Instance” window, keep the “Length” to 50 nm, and set the “width ” to 90 nm (Figure 11).
4. Place your transistor instance on the empty canvas in the “Virtuoso Schematic Editor” window.
5. Now you will place votage sources and ground. In the “Virtuoso Schematic Editor” window, go to Create → Instance or just press “i” from the keyboard. In the “Add Instance” Window, use “analogLib” for library, “vdc” for cell, “symbol” for view and “vlds” for Names (Figure 12). In the “Virtuoso Schematic Editor” window, place the instantiated DC voltage source. Similarly, instantiate a DC voltage source called “vvg” and place it in the “Virtuoso Schematic Editor” Window. Also instantiate a Ground called “gnd” and place it in the “Virtuoso Schematic Editor” window. We will use these two voltage sources and ground to do a DC sweep.

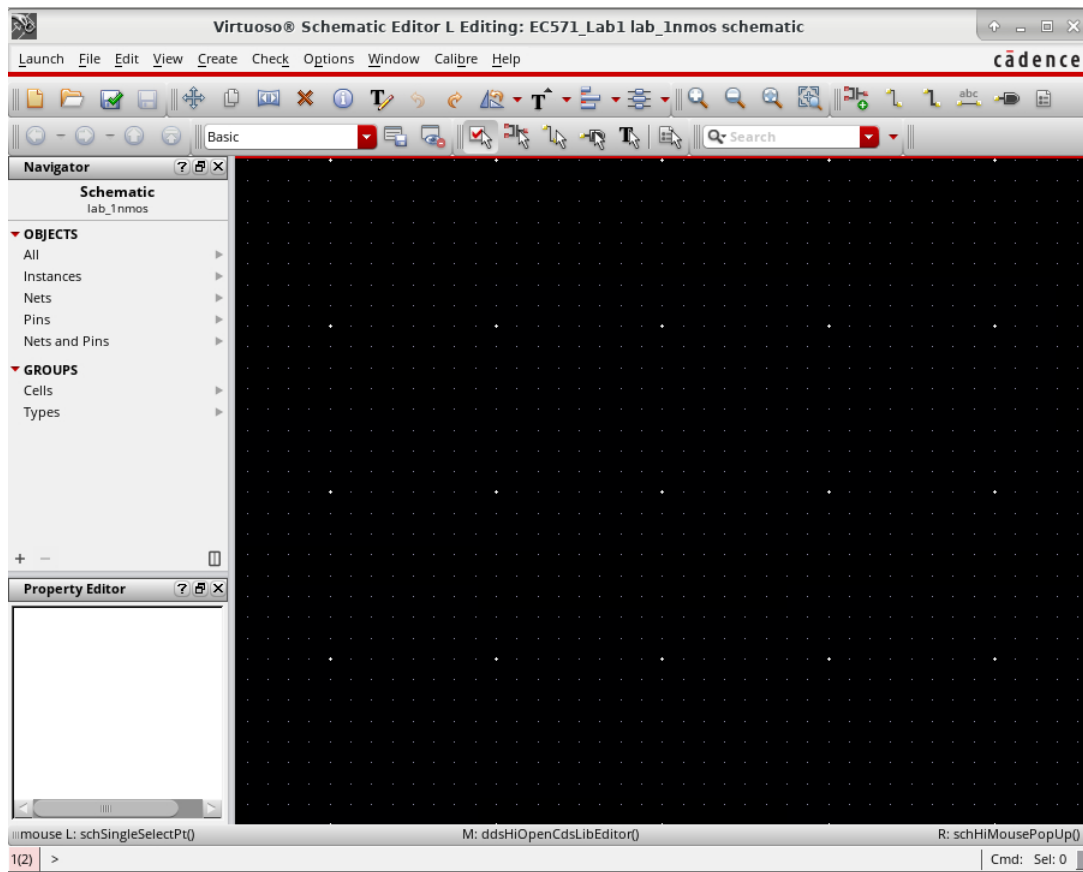


Figure 9: “Virtuoso Schematic Editor” window

6. Select the DC source with instance name “vlds”, press “q”. This will open the “Edit Object Properties” Window. Here set “DC voltage” to be “vlds”. Perform the same steps for “vlds” instance of the DC voltage source.
7. Your schematic after above steps should now look like Figure 10.
8. Once all the components are instantiated, wire them as shown in Figure 11. To connect the components click Create → Wire or just press “w” from the keyboard. You can use “ESC” to escape the wiring function.
9. Connect the bulk terminal of a NMOS (4th terminal between drain and source) to ground.
10. Label you drain and gate wires using “l” shortcut key. You can refer to other shortcuts in Figure 14.

Simulation Setup

1. We will use the the “Analog Design Environment” for simulation. For that click Launch → ADE L (“Analog Design Environment”) in the “Virtuoso Schematic Editor” window (Figure 15). If you get a warning, just click on “Yes”.

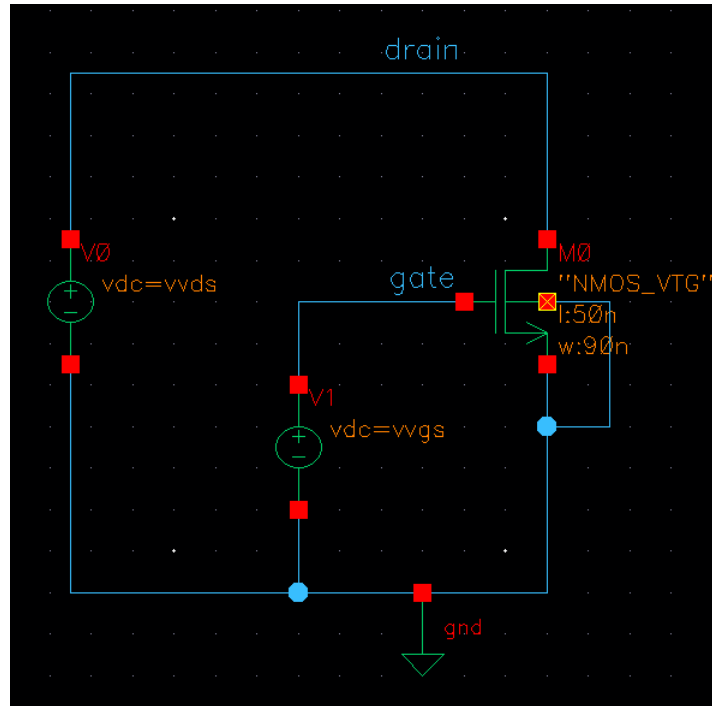


Figure 10: Schematic for NMOS DC sweep simulations

- Before running any simulations, the simulator needs to know about the process and the device models. In the “ADE L” Window, go to Setup → Model Libraries (Figure 16). This will open the “Model Library Setup” window.
- In the “Model Library Setup” window, click on “Click here to add model file” and click on “...”. In the window that opens, select “hspice_nom.include” file that is located under $\sim$ /EC571/FreePDK 45/custom_models. Click OK (Figure 17).
- In the “ADE L” Window, right click in the “Design Variables” panel and select “Copy From Cellview” (Figure 18). Cadence needs all variables initialized to some value, so set “vvds” = 1 and “vvgs” = 1 (see Figure 19) Cadence Virtuoso already knows that the units for vvds and vvg is volts.

DC Sweep Analysis (NMOS)

- You will do a DC sweep on the NMOS transistor to obtain I_{DS} vs. V_{GS} for a V_{DS} at 1 V.
- In the “ADE L” Window, go to Analyses → Choose (see Figure 20). This opens up a window called “Choosing Analyses”. Here choose dc analysis. DC analysis simulates a circuit’s steady-state (i.e. time-independent) behavior. Select “Save DC Operating Point”. Select Design Variable in the “Sweep Variable” panel. Click on the Select Design Variable and choose “vvgs”. In the “Sweep Range” panel, type 0 for start and 1 (this corresponds to V_{DD} voltage) for stop. For “Sweep Type”, select “Linear” then select “Step Size” and enter 0.1 in the field of “Step Size”. Select “Enabled” (see Figure 21). Click OK.
- To see the waveform, in the “ADE L” window, go to Output → To Be Saved → Select on Design (Figure 22). In the schematic window, click on the drain node of NMOS. In the “Outputs” panel

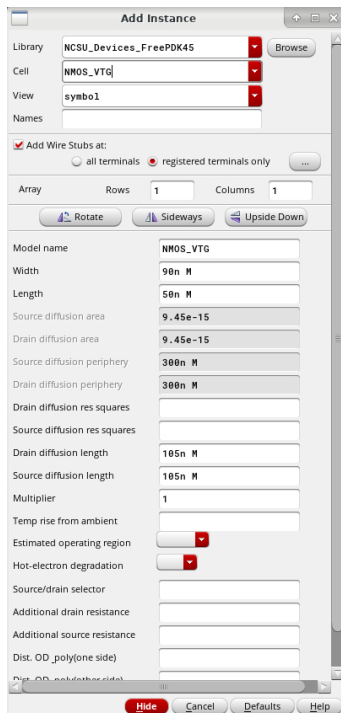


Figure 11: Adding instance view

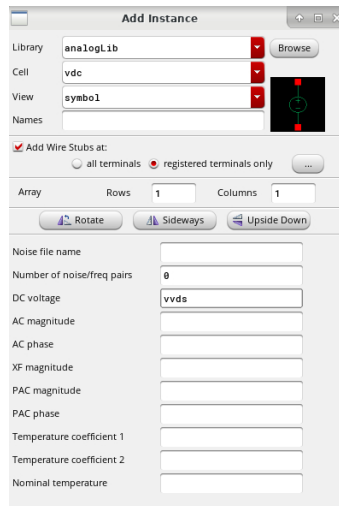
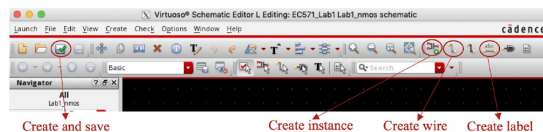


Figure 12: Adding voltage source view



Figure 13: Labeling wires



Keys	Function	Keys	Function
I	Add instance	Z	Zoom into selected region
W	Wiring	Ctrl + z	Zoom out selected region
M	Move	Ctrl + D	Deselect
R	Rotate	Shift + Q	Properties
F	Fit screen	L	Label
S	Stretch		

Figure 14: Most common shortcut keys

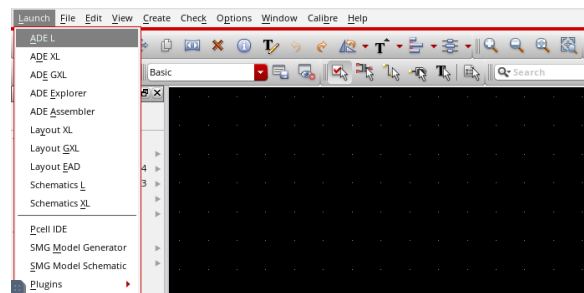


Figure 15: Launching “Analog Design Environment” (“ADE L”) from schematic view

of the “ADE L” window, you will see the selected outputs (see Figure 23). Remember to check “Plot” if you want to plot the desired waveform.

- If you have correctly completed all the steps so far, you will notice that there’s a circle highlighting the NMOS drain node in your schematic (see Figure 24). This means the simulator will plot current going through that node.
- In the “ADE L” Window, go to Simulation → Netlist and Run to run the simulation. Or, click the green button on the right-hand-side icon bar to run the simulation (see Figure 25). **If you change your schematic, remember to “Save and Check” again before you run the simulation. The simulation won’t run until you save the changed schematic.** The tool will give you an error in the case of unsaved schematic. You can check for your error or warning messages in you “CIW” window. If the simulation completed successfully, the “Visualization and Analysis XL” window will open where you will see a plot of I vs vvgs (see Figure 28)
- To print DC characteristics: in the “ADE L” window, click Results → Print → DC operating

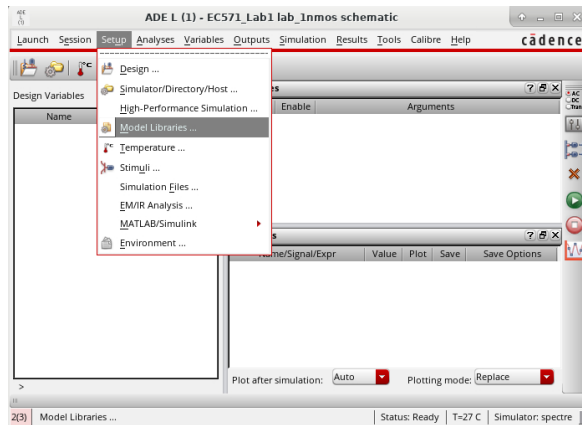


Figure 16: Opening “Model Library Setup”

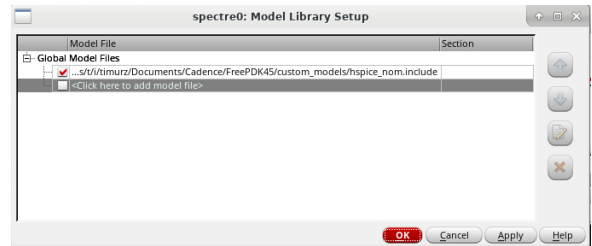


Figure 17: Setting up simulation models

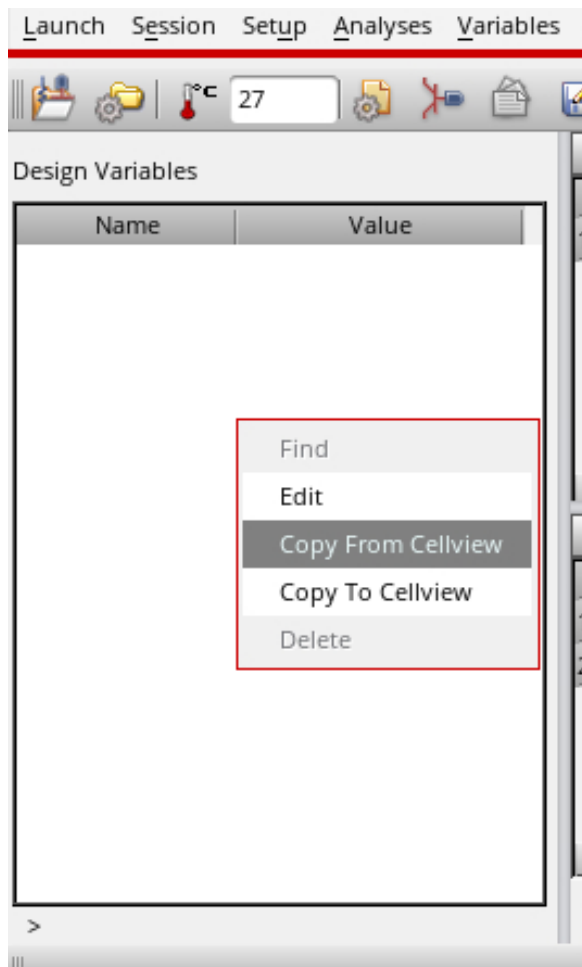


Figure 18: Copying variables from the schematic view

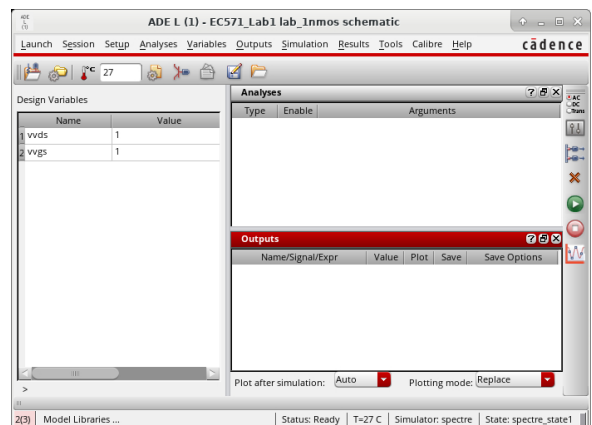


Figure 19: Setting up simulation variables in the “ADE L”

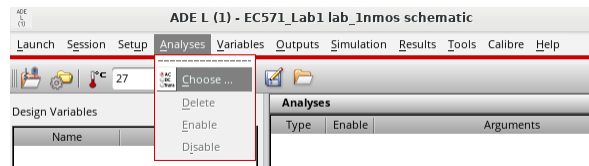


Figure 20: Opening analysis selection view

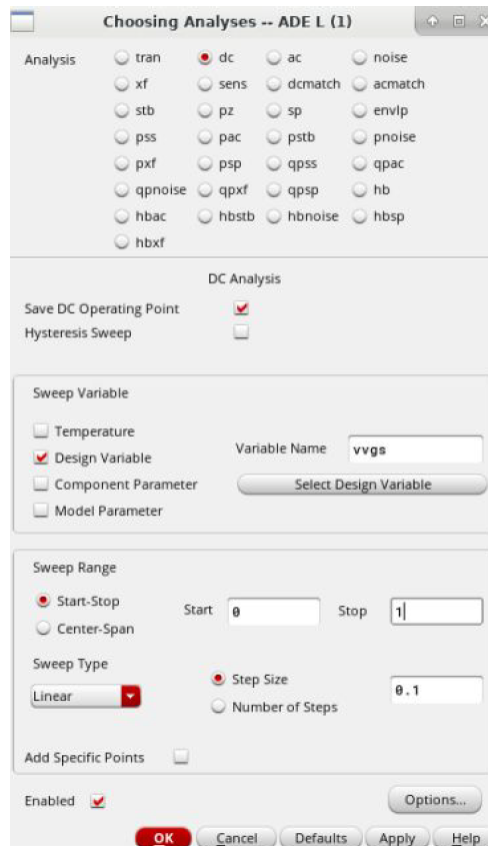


Figure 21: Setting up DC sweep simulation

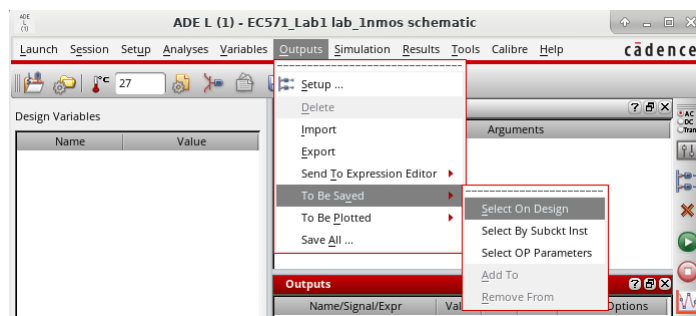


Figure 22: Opening output plot selection

points. Then select the NMOS component in the schematic window. The DC operating points will print out in the pop-up window. In the appeared list, search for id (drain current), region

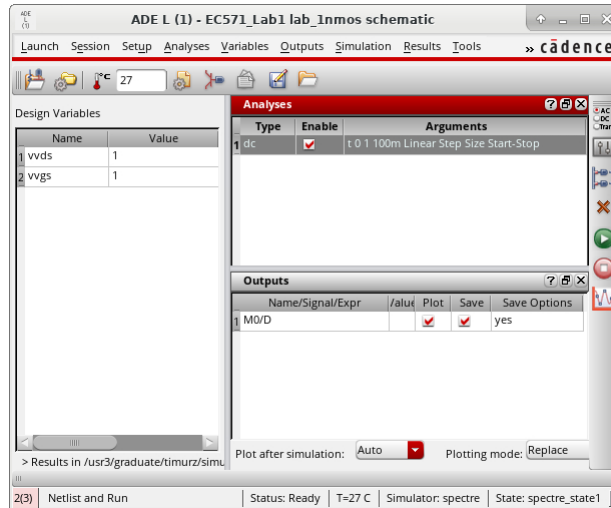


Figure 23: Setting up output plotting and saving in the “ADE L” window

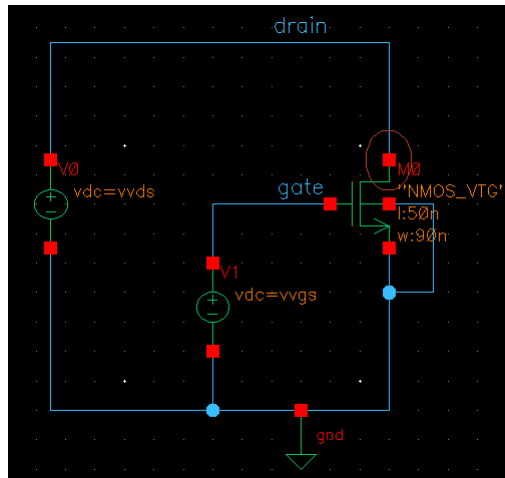


Figure 24: Schematic view with selected output

- (region of operation) values and other parameters such as v_{th} (threshold voltage), etc. The NMOS transistor is in saturation if region is 2 (see Figure 26).
7. In the “ADE L” Window, go to Session → Save State to save the current configuration of the simulation, so that you don’t have to reset the configuration next time. Save your state in the FreePDK45 folder. You can go to Session → Load State to load saved states.
 8. The output of a simulation is plotted when the simulation ends. You can also plot it by going to Results → Plot outputs → DC, or by simply clicking on small graph button on right in the “ADE L” window as shows on Figure 27.
 9. The output should look like the one shown in Figure 28 (you can change the properties of the graph by right clicking on the graph).

Running I_{DS} vs. V_{DS} : for different values of V_{GS}

In this part you will simulate and plot I_{DS} vs. V_{DS} curves for different values of V_{GS} .

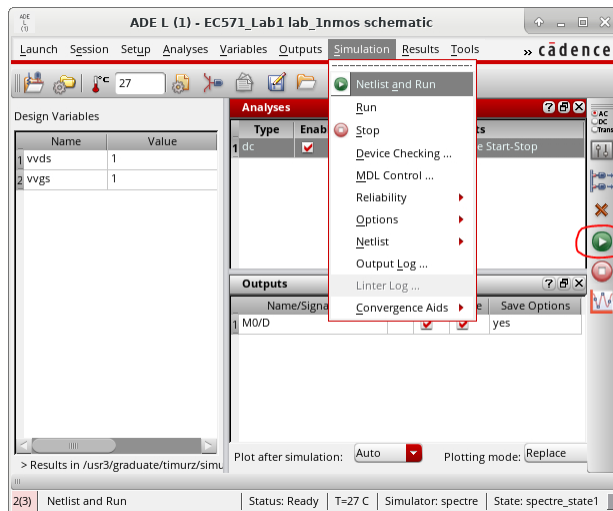


Figure 25: Running the simulation from “ADE L” window

Parameter	Value
gbd	1p
gbs	1p
gds	17.19u
gmb	36.33u
gmbs	36.33u
gmoverid	1.534
ibe	-122.9p
ibulk	-121.9p
id	81.68u
idb	1p
ide	81.68u
ids	81.68u
igb	121.6p
igbacc	121.6p
igbinv	121.6p
igcd	38.84p
igcs	54.06p
igd	-4.412y
igdt	258.7p
ige	258.7p
igidl	0
igisl	0
igs	44.21p
isb	-229.6y
ise	-81.68u
isub	292.2f
pwr	81.68u
qb	-25.07a
qbi	-25.07a
qd	-9.439a
qdi	-9.439a
qg	67.39a
qgdov1	1.743a
qgi	67.39a
qgsav1	21.35a
qjd	-136.2a
qjs	31.37e-27
qs	-32.88a
qsi	-32.88a
rdeff	0
region	2
reversed	0
rgate	240m
rgbd	0
ron	12.24K
rout	58.16K
rseff	0

Figure 26: Printed DC operating point data

1. Set up DC analysis as before, but instead select “vvd” in “Variable Name” (Figure 29).
2. To sweep V_{GS} , in the “ADE L” window, go to Tools → Parametric Analysis. Set up parameters for “vvgS” as in the Figure 30. Click the green button “Run Selected Sweep” to run the parametric simulation.

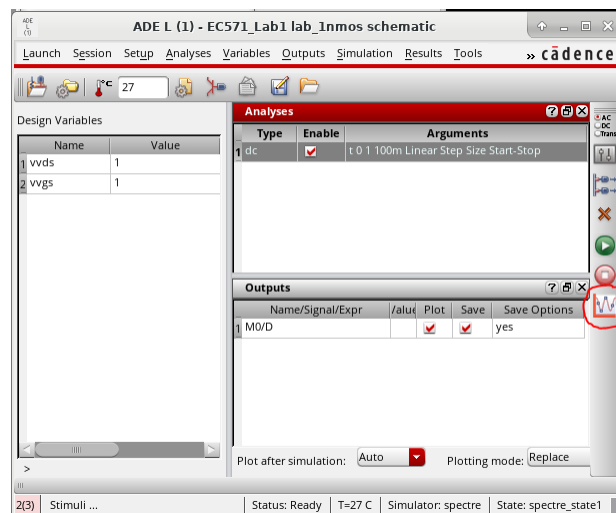


Figure 27: Plot your simulation results from the “ADE L”

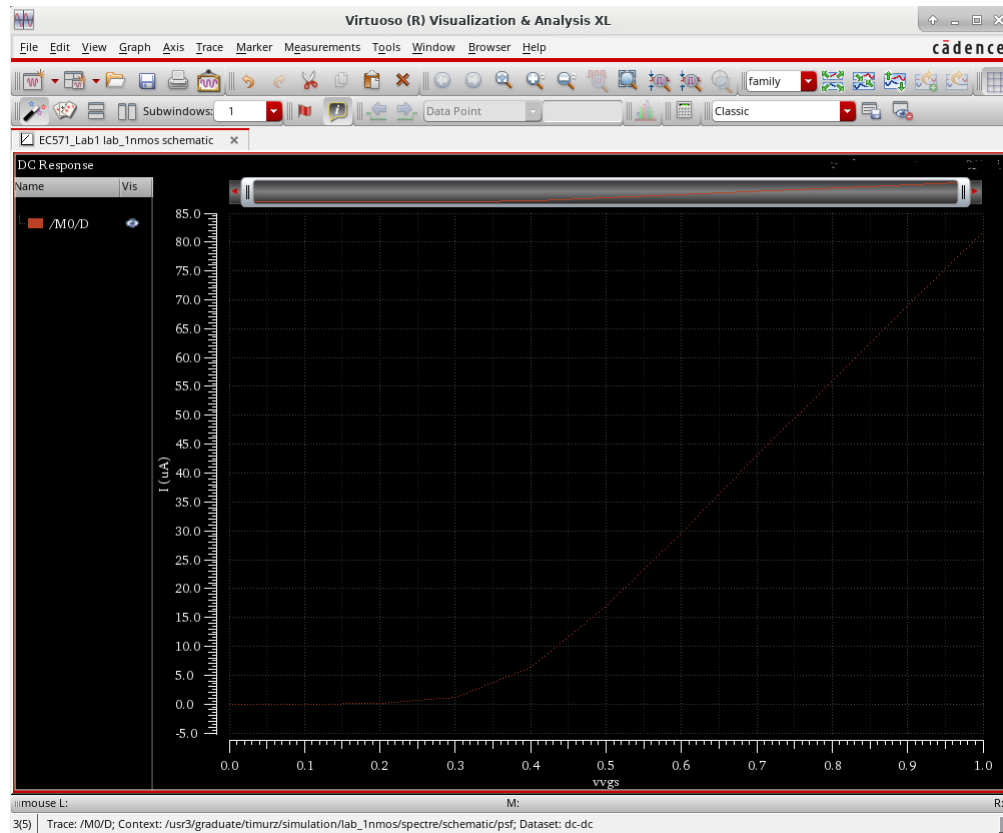


Figure 28: I_D vs. V_{GS} for NMOS

3. The resulting plot will look like in Figure 31.

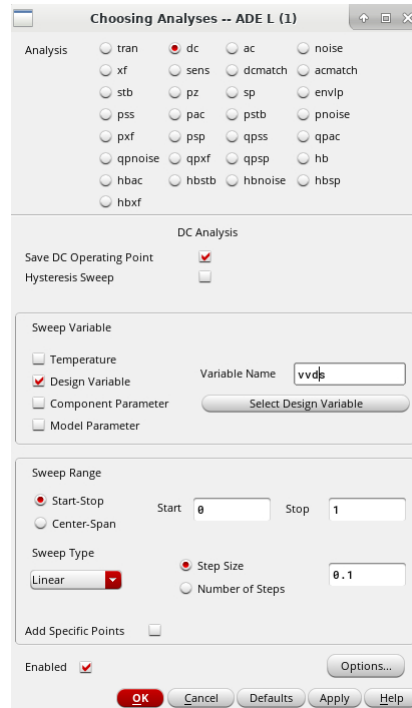


Figure 29: Changing sweep variable to “vvd”

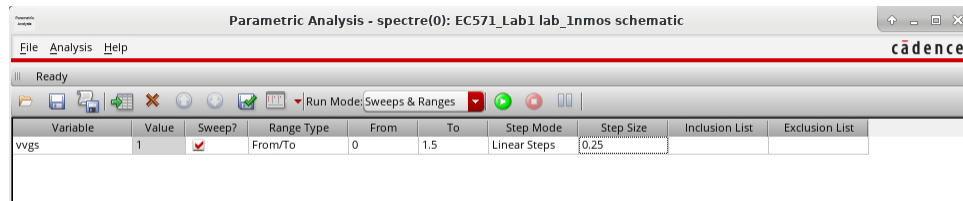


Figure 30: Setting up parametric analysis

PMOS Characterization Analysis

1. The schematic of a PMOS circuit is shown in Figure 32 . The bulk is connected to V_{DD} instead of GND.
2. You have to set up you DC simulation similar to NMOS case, as shown in Figure 33. Note that I_{SD} is selected from the source (bottom) port this time.
3. Run your simulations
4. Simulation results will look like as in Figure 34.
5. Now, you have to setup the parametric simulations to obtain I_{SD} vs. V_{SD} graph for different V_{SG} values, similar to NMOS simulations above. Compare your results with the one in Figure 35.

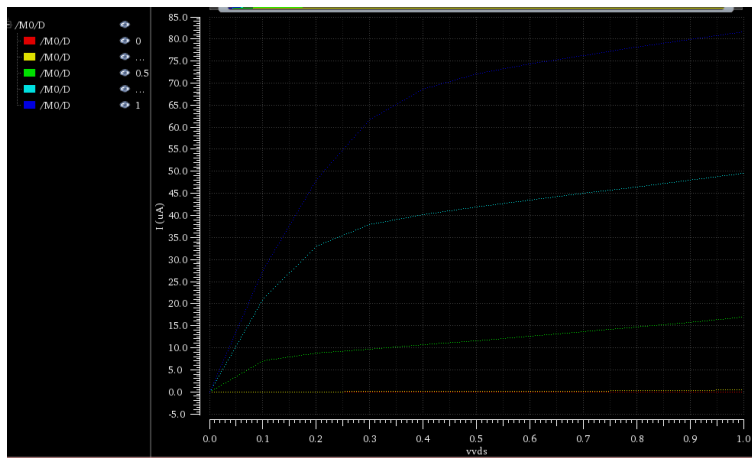


Figure 31: Results of parametric simulation for NMOS

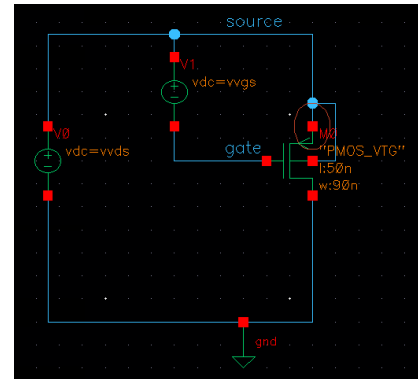


Figure 32: PMOS schematic

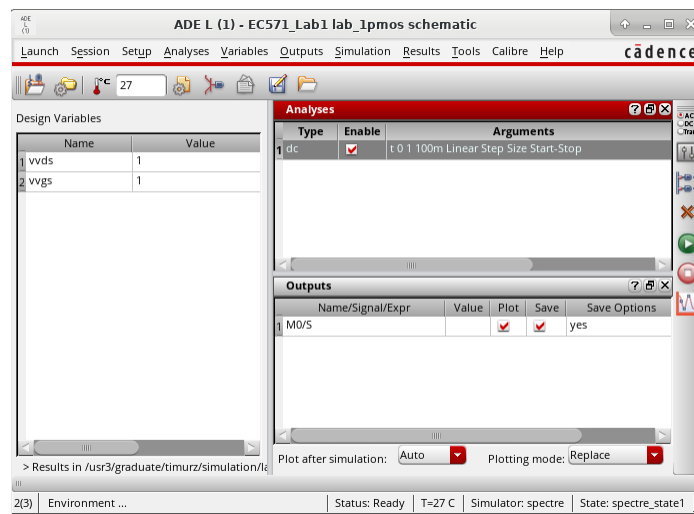
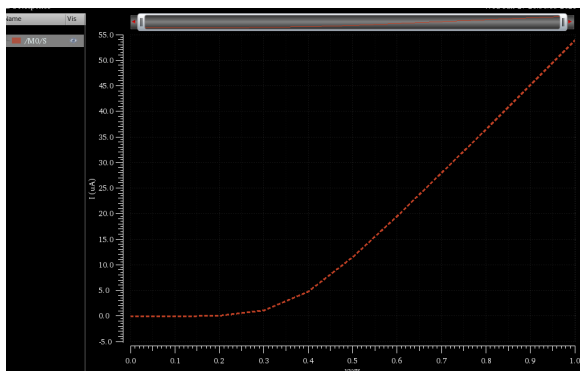
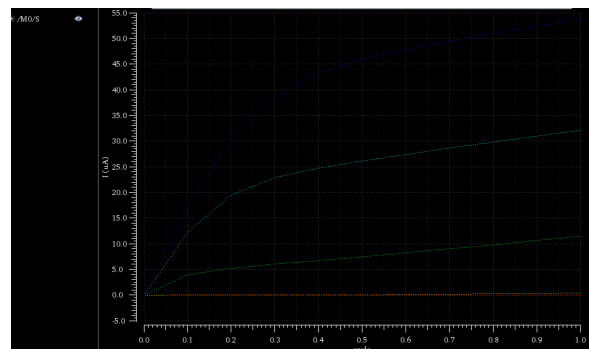
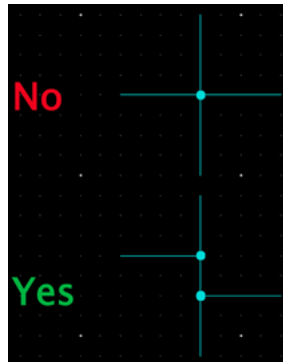
Figure 33: The “ADE L” window setup for I_{SD} vs. V_{SG} curve simulationsFigure 34: I_{SD} vs. V_{SG} for PMOS

Figure 35: Results of parametric simulation for PMOS

Guidelines for Schematic Editing

- Naming and Wiring
 - Avoid global names (such as “gnd!”.)
 - Use wires to connect nodes and elements, do not make connections by naming.
 - Avoid cross connection.



- Transistors Symbols
 - Only use 4-terminal MOS symbols.

This forces you to think of where to substrate should be connected.

- Power Supply and GND
 - Do not use the special VDD symbols.
 - Do not use the special GND symbols.
 - Only use one node connected to GND so the simulator has a reference node.

Power supply and ground nodes are connected to many components in the schematic, which leaves them hard to trace. Avoid having multiple power supply and ground nodes will reduce the complexity of the schematic.

- Pin Placement
 - Keep the positive supply rail at the top of the schematic, the negative supply rail at the bottom of the schematic.
 - Keep all input, control and bias pins lined up at the left side of the schematic, all output pins at the right.
- Signal Flow
 - Keep the signals going from the left to the right side of the schematic.